Satellite-Ground Integrated External Calibration of the WindRAD Scatterometer Onboard FY-3E Satellite

Jian Shang, Haoqiang Shi, Mei Yuan, Qiong Wu, Ailing Lv, Fangli Dou, Honggang Yin, Lixia Liu, and Xiuqing Hu

***Abstract*— The first scatterometer onboard Chinese meteorological satellites is a dual-frequency and dual-polarization scatterometer named Wind Radar (WindRAD). It uses an advanced fan-beam conical scanning mechanism to acquire wind vector observation of global ocean surfaces and other geophysical parameters. Since the launch in 2021, external calibration using ground-based active radar calibrator (ARC) has been carried out to evaluated WindRAD in-orbit situation and observation accuracy. Satellite-ground integrated external calibration process is proposed, and optimal satellite-ground observation mode is specially designed for the WindRAD, greatly reducing the complexity of satellite-ground interaction and improving the efficiency of external calibration observation. This article proposes the algorithm of WindRAD external calibration, with comprehensive consideration of scientific nature and engineering realizability. WindRAD in-orbit observation data as well as ARC data were used to calculate the real antenna patterns and absolute calibration coefficients of each polarization for both C- and Ku-bands. The evaluation results revealed that the in-orbit antenna pattern hadn't changed much compared with the prelaunch test result, which for the first time confirmed the correctness of the key parameters used in WindRAD calibration. Moreover, the calibration accuracy is better than 1 dB. For the first time, the stability of WindRAD in orbit is confirmed using external active reference target.**

***Index Terms*— Active radar calibrator (ARC), backscattering coefficient, external calibration, FengYun-3E (FY-3E) satellite, scatterometer, wind radar (WindRAD).**

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I. Introduction

SPACEBORNE scatterometers are active microwave radar systems designed to provide global backscatter measurements [1], [2], [3], [4]. The latest spaceborne scatterometer, Wind Radar (WindRAD), was launched in 2021 onboard the FengYun-3E (FY-3E) meteorological satellite [5], [6], which is the first active remote sensing instrument of Chinese FengYun series satellite. WindRAD is also the first dual-frequency scatterometer in the world operating at C- and Ku-bands, both with horizontal and vertical polarization measurements, providing unique information for product retrieval. Furthermore, the advanced rotating fan-beam conical scanning mode makes it possible to obtain abundant observing geometry. The WindRAD is mainly aimed at measuring wind vector over global ocean surfaces, including wind speed and wind direction, all weather and all day with high precision as well as high resolution [7]. In addition, the WindRAD can also measure soil moisture, sea ice, and other geophysical parameters. The WindRAD was turned on and has been conducting global observations since July 9, 2021. During the in-orbit test, calibration and validation were performed, and its operating condition is rather steady [8].

The WindRAD performs both internal calibration and external calibration to ensure the measurement accuracy in orbit. The internal calibration is performed twice for every conical scan of the WindRAD [9]. A portion of the emitted energy is fed back to the receiver, which provides a valuable evaluation of the transmitting power and receiving gain. The evaluation value is used for the calculation of the normalized radar cross section (NRCS, or σ^0). Because the internal calibration signal does not pass through the radar transceiver front end to the antenna part, it is impossible to correct the parameter change of the whole radar, so it is necessary to calibrate the whole radar absolutely by external calibration. The external calibration absolutely corrects the radar observation of wind field measurement through the calibration target on the ground, so as to eliminate some difficult-to-measure and difficult-to-estimate quantities that are not taken into account in the antenna pattern and radar equation. The realization methods of external calibration can be divided into passive method and active method. The passive point target method uses standard reference reflector for calibration, which requires the reflector size to be more than 10 m, making it difficult to process with high precision. The ground distributed target method uses a

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large area of the ground to calibrate the target with stable scattering, among which the tropical rain forest is an ideal calibration target [10], [11], [12], [13], [14]. Studies have shown that the calibration accuracy using the Amazon rain forest may reach about 1 dB [15], [16], but the backscattering of the tropical rain forest changes with the seasons. Therefore, these passive methods cannot meet at least 1 dB external calibration requirements of the WindRAD. Active calibration method uses active radar calibrator (ARC), which can process the original radar signal with time delay and frequency shift and has the advantages of compact structure, wide antenna beam, large radar cross section (RCS), stability, and high accuracy. The active microwave instrumentation (AMI) scatterometer carried on the European remote sensing (ERS-1) and (ERS-2) satellite was calibrated using three methods, active calibrator calibration, Amazon rainforest calibration, and ocean target calibration. The calibration results indicated that the combined use of the three methods shows a good performance, meeting the requirement for calibration accuracy of 0.7 dB [17], [18], [19]. Similar with the calibration of the ERS scatterometers, the NASA scatterometer (NSCAT) onboard the advanced Earth observing satellite (ADEOS) [20], [21], [22], and the advanced scatterometer (ASCAT) on the MetOp satellite [23], [24] also employed the combination method of the three calibration techniques. The Seawinds and QSCAT scatterometers onboard ADEOS-2 and QuikSCAT satellites, respectively, were calibrated using passive calibrators, whose results met the calibration accuracy of 0.9 dB [25], [26]. The WindRAD adopts active calibration means using the ARC, numerical weather prediction Ocean calibration (NOC), and rainforest validation.

In this article, the satellite-ground integrated external calibration using the ARC for the WindRAD is discussed. This article is organized as follows. Section II introduces the external calibration process designed for the WindRAD and optimal design of satellite-ground observation mode. Section III presents data processing algorithm of the external calibration. Section IV is dedicated to the preliminary results of external calibration. Discussions and conclusions are included in Section V.

II. External Calibration Technique for the WindRAD

A. WindRAD Overview

The WindRAD onboard FY-3E satellite carries out conical scan with fan beams, and the antenna rotates 360° without interruption at a speed of 0.6 rd/s. The viewing geometry of WindRAD is shown in Fig. 1 [8]. The main operating modes of the WindRAD include an echo measurement mode and an external calibration mode. The former is the default operating mode, in which period, the Earth observations, internal calibrations, and noise measurements are performed sequentially, according to the designed pulse repetition time. The latter requires sending remote control commands to the instrument as well as the ground-based equipment ARC.

The ARC needs to have the following main functions.

1) *Receiving Function:* It can receive the transmitted signal from the WindRAD.

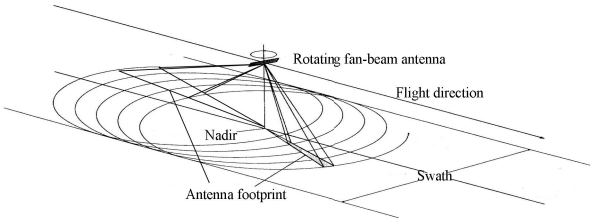


Fig. 1. Viewing geometry of WindRAD.

TABLE I
Main Specifications of the ARC

	Item	Specification	
		C band	Ku band
Spectrum specifications	Central frequency	5.4 GHz	13.256 GHz
	Polarization	V、H	V、H
	Polarization isolation	⩾ 30 dB	⩾ 30 dB
Geometry specifications	3dB beam width	≥ 10°	
	Accuracy of beam pointing control	≤ 0.5°	
	Accuracy of beam pointing measurement	≤ 0.1°	
Radiometric specifications	NRCS	⩾ 90 dBm^2	
	Stability of NRCS	≤ 0.15 dB/10min	
	Uncertainty of NRCS	≤ 0.25 dB	
	Absolute calibration	≤ 0.3 dB	

2) *Transmitting Function:* It is capable of transmitting signals that can be received by the WindRAD.
3) *Time-Delay/Frequency-Shift Transfer Function:* It can transmit the received signals transmitted by the WindRAD to the WindRAD after delay (or frequency shift).
4) *Automatic Tracking Function:* It can automatically determine its position information (latitude and longitude, altitude), attitude information (azimuth and horizontal attitude), and time information, and can automatically aim at the designated satellite.
5) *Scanning Function:* It can scan in 360 omni-directional and 90 pitch angle range.
6) *Remote Control Function:* It can be remotely controlled from the data processing center.
7) *Monitoring Function:* It can automatically monitor its key state parameters.
8) *Self-Calibration Function:* It can effectively calibrate its own measurement accuracy.
9) *Preprocessing Function:* It can analyze the received signal preliminary and screen out valid data to greatly reduce the amount of data that need to be sent back to the data processing center.

The main specifications of the ARC developed for the FY-3E/WindRAD is listed in Table I. It works at the frequencies of 5.4 and 13.256 GHz, and has two polarizations, which are the same as the WindRAD. In order to ensure that the main beam can effectively capture the WindRAD, the 3 dB beamwidth of the ARC is wider than 10°. The antenna pointing measurement accuracy is better than 0.1°. As the ARC is considered to be a reference to the WindRAD, the absolute calibration accuracy of the ARC itself is better than 0.3 dB.

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As mentioned before, ARC is designed to have three working modes, receiving mode, transmitting mode, and transfer mode.

1) *ARC Receiving Mode:* The WindRAD transmits a pulse signal and the ARC receives the signal. The actual WindRAD transmitting antenna pattern can be obtained through data processing.
2) *ARC Transmitting Mode:* The ARC transmits the specified signal to the spaceborne WindRAD and the WindRAD receives the signal. The actual WindRAD receiving antenna pattern can be obtained through data processing. Because the WindRAD antenna is passive, the pattern of the transmitting antenna is the same as that of the receiving antenna, and the ARC does not need to work in the transmitting mode in practical use.
3) *ARC Transfer Mode:* The WindRAD transmits a pulse signal and the ARC receives the signal. Then the received signal is transferred after time delay/frequency shift and amplification, and is received by the WindRAD. After processing the valid data, the absolute calibration coefficient of the WindRAD can be calculated.

B. WindRAD External Calibration Process

The complete satellite-ground integrated external calibration workflow for the WindRAD is designed as follows. When the WindRAD needs to carry out external calibration, a task schedule is generated in the ground system and transferred to the external calibration station, including the timetable, work mode, time delay/frequency shift parameter, attenuation settings for transmitting channels and receiving channels, and accurate pointing angles. The ARC and ancillary equipment will open automatically and complete preheating and self-inspection in advance. The antenna orientation of the ARC is set according to the pointing angles calculated using the forecast satellite orbit. On the other side, a remote control instruction is sent to the satellite through the ground control center to switch the WindRAD from the default operating mode to the external calibration mode on time. Then the ARC and the WindRAD work together during the overhead period. When the WindRAD transmits signals, the ARC on the ground receives the signal and stores its amplitude. If the ARC retransmits the signal back, the WindRAD can receive it and the raw data will be downlinked to the ground station. After this stage, another remote control instruction is sent to the satellite to switch the WindRAD back to the default operating mode, as shown in Fig. 2. The ARC and ancillary equipment on the ground will turn off automatically according to the task schedule. The synchronous observation data from the external calibration station is preliminarily analyzed and only the valid data will be transmitted to the data processing center for further processing. To summarize the external calibration mode, the automatic gain control (AGC) value of the WindRAD is changed in order to detect the ARC signal, and the interruption to WindRAD default operating mode is about 10 min.

There are several particularities in both ARC design as well as external calibration data processing. First, in order to make the signals forwarded by the ARC fall outside the frequency band of the WindRAD 3 dB beam, which can improve the signal-to-noise ratio of external calibration, the ARC has added frequency shift design. Second, the WindRAD adopts advanced fan-beam conical scanning system, whose beamwidth in the range-direction is relatively wider than typical scatterometers in orbit like HY-2 scatterometers. Thus to completely cover the wide beam, new method can be used to adjust different beam directions of the ARC to obtained the range-direction pattern of the WindRAD. Third, the Ku-band azimuth beam of the WindRAD is narrow (1.5° for the HY-2 scatterometers and 0.8° for FY-3E WindRAD). In order to obtain the accurate beam pointing information of the WindRAD and improve the calculation accuracy during external calibration, refined processing has been carried out in the calculation of the scanning angle of the WindRAD, including finding the optimal angle by matching the changes in the received pulse power with the antenna pattern.

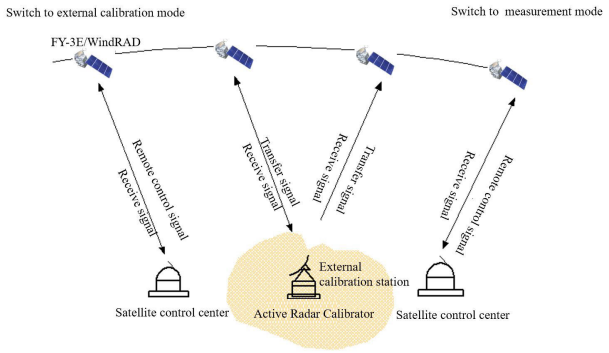


Fig. 2. Switching between WindRAD default operating mode and external calibration mode.

Based on the proposed external calibration process, the workflow is shown in Fig. 3.

C. Optimal Design of Satellite-Ground Observation Mode

According to the originally designed scheme, the external calibration process of the WindRAD needs to upload external calibration work instructions and switch the working mode of the WindRAD. The process is complicated and the frequency of implementation is quite limited. In order to reduce the complexity and improve the efficiency of the external calibration, it is necessary to adapt to the WindRAD AGC value by adjusting the working power parameters of the ARC. In order to achieve this goal, and at the same time, meet the ARC RCS quality requirements, the signal power of the ARC should be less than the saturation power of the on-board receiver and the analog-to-digital (AD) device, the attenuation parameters of the ARC channel are optimized and adjusted for many times, and the new technology of frequency shift forwarding signal is also used so that the on-board radar can complete the external calibration without switching the working mode, as shown in Fig. 4, eliminating the links of instruction upload and manual intervention. The complexity and uncontrollable factors caused by the operation of switching modes are avoided, the complexity of satellite-ground interaction is greatly reduced, and the occasional external calibration task is realized as a daily observation task, which greatly increases the frequency of external calibration. Also, it is not necessary to rudely

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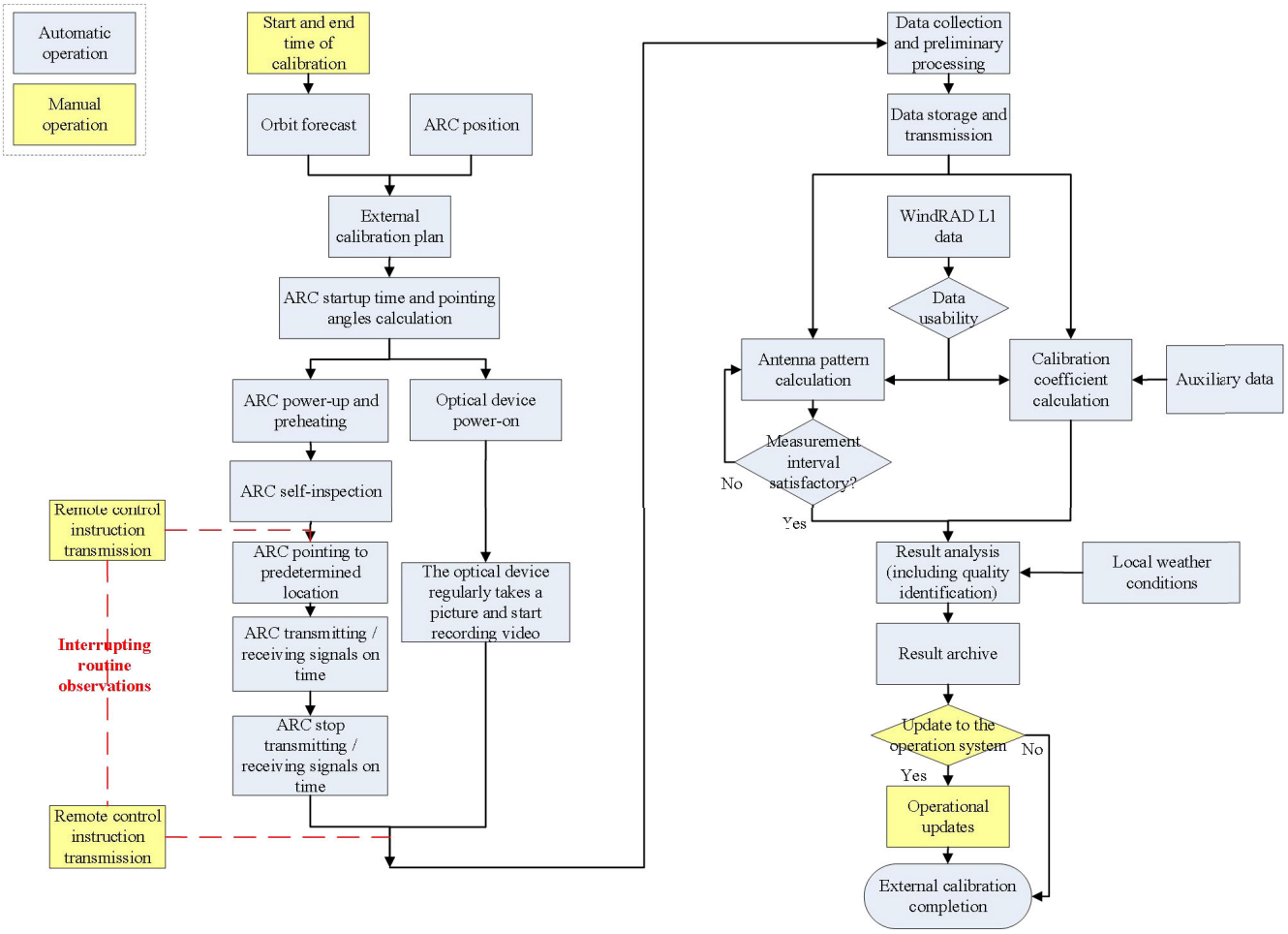


Fig. 3. Complete workflow of satellite-ground integrated external calibration for the WindRAD.

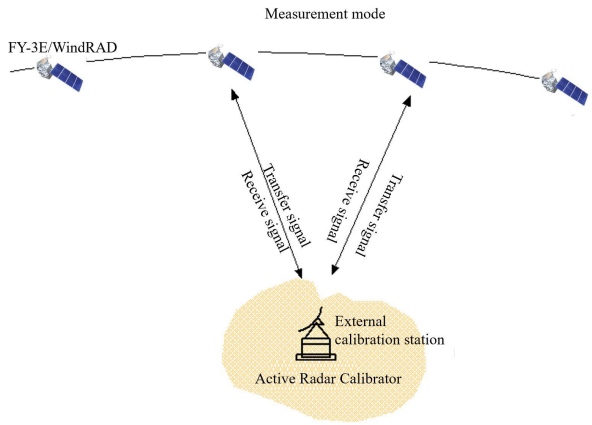


Fig. 4. Optimized WindRAD external calibration mode.

interrupt the routine observation of the WindRAD, providing the public users with almost continuous observation data, with only a few seconds of data for internal use.

III. Algorithm of WindRAD External Calibration

WindRAD raw data are received and processed by the National Satellite Meteorological Center, China Meteorological Administration (NSMC, CMA), which is the data processing center of Fengyun meteorological satellites. As previously mentioned, according to the proposed optimal satellite-ground observation mode, the external calibration for the WindRAD can be carried out almost every day, as long as the satellite flies over the external calibration station, other than restricted times like half a year or in case of changed data quality; thus, plenty of external calibration data should be processed automatically. The flow diagram of the external calibration algorithm proposed for the WindRAD is shown in Fig. 5.

First, data unpacking and quality control were performed on level 0 (L0) data to obtain level 1A (L1A) data. According to the task schedule, the L1A data corresponding to the external calibration is extracted from the whole L1A data.

Second, geolocation was conducted, using the satellite orbit, attitude, observation time, latitude, longitude, pointing angles, and time delay of the ARC. The antenna intersection angles both in the WindRAD coordinate system and in the ARC coordinate system are calculated using the line of sight (LOS) and coordinate system transformations. Due to the existence of angular errors introduced by scanning azimuth, the attitude, and so on, there are certain errors in the antenna pitch angle and azimuth angle of each pulse calculated by the satellite-ground geometric relationship. This error will affect the measurement accuracy of the in-orbit directional pattern

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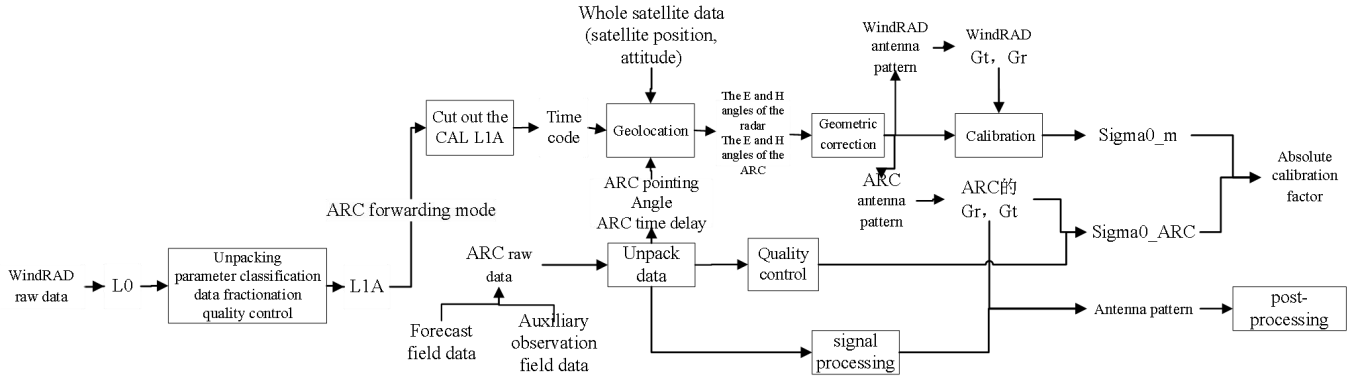


Fig. 5. Flow diagram of WindRAD external calibration algorithm.

as well as the measurement accuracy of the absolute calibration coefficient. Therefore, geometric correction is needed to correct this deviation, which is an important procedure in the external calibration. Establish the objective equation f as

$$f(\Delta\theta, \Delta\varphi) = \frac{1}{N}\sum_{i=1}^{N}(G_{\text{meas}}(\theta_m(i) + \Delta\theta, \varphi_m(i) + \Delta\varphi) - G_{\text{ground}}(\theta_m(i), \varphi_m(i)))^2 \quad (1)$$

where G_{meas} is the directional pattern measured by the ARC, G_{ground} is the directional pattern of the WindRAD in ground testing. $\Delta\theta$ and $\Delta\varphi$ are the pitch angle and azimuth angle deviations caused by various angular deviations, respectively, and the value range is between $\pm 0.5°$. θ_m and $\Delta\varphi$ are the range and azimuth angles in the WindRAD antenna coordinate system obtained according to the satellite-ground geometry relationship. N is the number of sampling samples used for calibration processing. The values of $\Delta\theta$ and $\Delta\varphi$ that satisfy the minimum value of the objective function are the estimated values, then the antenna gains of the WindRAD and the ARC are calculated based on the accurate antenna intersection angles.

Third, the data received by the ARC is unpacked and strict quality control is required for the observation data. Determine the data usability according to the weather conditions and eliminate the data that are greatly affected by the cloud, rain, and atmosphere to avoid the impact of attenuation. Reject invalid data according to the signal strength to avoid affecting the subsequent data processing. Furthermore, for a single flyover data processing result to be mentioned later, the discreteness of the result should be judged according to the historical data, and the processing result with large fluctuation should also be removed.

Four, complicated signal processing is executed to obtain the radar signal amplitude. IQ demodulation of the original data SigRF is carried out using the following equation:

$$\begin{aligned} I &= \text{SigRF} * \cos(2 * \text{pi} * fc * t) \\ Q &= \text{SigRF} * \sin(2 * \text{pi} * fc * t) \end{aligned} \quad (2)$$

where fc is the intermediate frequency and t is the time of each sampling. After IQ demodulation, complex data are formed. The ARC adopts intermediate frequency sampling, and the signal needs to be down-converted to obtain baseband waveform. The filter window should be constructed according to the number of samplings and signal bandwidth

$$N_lvb = \text{round}(1.1 * \text{Br}/\text{Fs} * L_\text{sample}) \quad (3)$$

where Br is the bandwidth of the WindRAD transmission signal, Fs is the sampling frequency of the ARC, and L_sample is the number of samplings. The round() means round up or down to the results. After conversion to the baseband, downsampling can be carried out. Then perform matched filtering on the complex signal to obtain a matched filtered waveform. Pulse compression performs matched filtering on each received linear frequency modulation (LFM) pulse signal $\text{Sig}(t)$, wherein the matched filter is

$$H(t) = \text{Conj}(\text{Sig}(-t)). \quad (4)$$

And a matching function is constructed

$$H(t) = \exp(-j * \text{pi} * \text{Kr} * t \wedge 2) \quad (5)$$

where Kr is the FM slope. The pulse compression mode is

$$\text{SigRF_Out} = \text{ifft}(\text{fft}(\text{Sig}(t)) * \text{fft}(H(t))). \quad (6)$$

Then calculate the signal amplitude of the complex signal after pulse compression. According to the set threshold, extract the effective pulse signal and thus obtain the sampling of the distance-oriented tangent plane during a single crossing of the WindRAD.

The fifth step is to fit the antenna pattern. As the beamwidth of the WindRAD in the distance direction is wide, which is quite different from the pencil beam often used by the scatterometers, every time the satellite flies over the ARC, the ARC can only receive some sampling data of the WindRAD. Therefore, it is necessary to fit the WindRAD antenna pattern by combining the aforementioned processing results of multiple flyovers. In order to accumulate adequate measurement data that meet the requirement of fitting with an interval of $0.1°$ for the antenna pattern, it takes approximately four orbital

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regression cycles of the satellite, which is about 22 days. Considering the weather factors such as rainfall and snowfall, the time required is generally 1–2 months.

According to the above algorithm, the real antenna pattern is obtained, which lays the foundation for the in-orbit detection and real-time correction of the antenna pattern for the WindRAD.

Another function of the external calibration is to calculate the absolute calibration coefficient, which algorithm is also developed of the WindRAD. To the WindRAD onboard the satellite, the ARC appears like a target with maximum RCS $\mathrm{RCS}_{\mathrm{theo}}$ given by

$$\mathrm{RCS}_{\mathrm{theo}} = \frac{\lambda^2}{4\pi} * G_{r\mathrm{ARC}} * G_a * G_{t\mathrm{ARC}} \tag{7}$$

where $G_{t\mathrm{ARC}}$ and $G_{r\mathrm{ARC}}$ are the gains of the ARC transmit and receive antennas, and G_a is the loop gain, which includes the gain of the radio frequency amplifier as well as any mismatch or transmission line losses.

The WindRAD calibration coefficient k can be estimated from $\mathrm{RCS}_{\mathrm{theo}}$ and the RCS measured by WindRAD $\mathrm{RCS}_{\mathrm{meas}}$

$$k = \frac{\mathrm{RCS}_{\mathrm{meas}}}{\mathrm{RCS}_{\mathrm{theo}}} \tag{8}$$

$$\mathrm{RCS}_{\mathrm{meas}} = \frac{P_s(4\pi)^3 R^4 L_w L_{\mathrm{cal}}}{\lambda^2 G_r G_t P_{\mathrm{cal}} L_f L_{sc}} \tag{9}$$

where P_s is the WindRAD received power of the return signal at normal mode, R is the range from the ARC to WindRAD, L_w is the total insertion loss of the radar system, L_{cal} and L_{sc} represent the receiver attenuation in calibration mode and normal mode, G_r and G_t are the gains of the WindRAD transmit and receive antennas, P_{cal} is the WindRAD received power of the return signal at calibration mode, L_f represents the loss in internal calibration units when operating at calibrating mode.

IV. Preliminary Results of WindRAD External Calibration

A. External Calibration Station

The first external calibration station of Fengyun satellites has been built in Xilinhot National Climate Observatory, Inner Mongolia, China. The underlying surface of the station is grassland, far away from urban areas, thus, the NRCS of the ground surface is small, which can effectively suppress the influence of ground echo. The terrain changes gently, avoiding the sudden increase of ground backscattering, and there is no obstruction of terrain, buildings and vegetation, so as to ensure the normal transmission and reception of radar signals. Also, there is no obvious electromagnetic interference within the working frequency range of the ARC. In terms of climatic conditions, drought and lack of rain can reduce the influence of atmosphere on electromagnetic wave attenuation and improve the accuracy of external calibration. The landform and appearance of the Xilinhot external calibration station are shown in Fig. 6.

Since the postlaunch testing phase, the ARC has been put in to use to evaluation the in-orbit antenna pattern for the WindRAD. After the optimization of the satellite-ground observation mode of WindRAD external calibration, the task changed from initial experimental task to normal operation. According to the results of satellite orbit prediction, if the satellite flies across the ARC, it will automatically start the external calibration task, thus accumulating a large number of actual observation data, which can be used to analyze the changes of antenna pattern in orbit and also calculate the absolute calibration coefficient. The data from November 2023 to January 2024 are used for analysis in this article, including the WindRAD observation data, ARC data, and other auxiliary data.



Fig. 6. Xilinhot external calibration station.

B. Results of In-Orbit Antenna Pattern

Using the algorithm proposed in Section III, the in-orbit antenna patterns for each flyover are calculated for C- and Ku-bands, including both HH and VV polarizations, shown in Fig. 7. Fig. 7(a) and (b) are the antenna patterns of HH and VV polarizations for C-band, respectively; (c) and (d) are those of HH and VV polarizations for Ku-band, respectively. For different flyovers of the WindRAD, the observation geometries are different and thus 1-D antenna pattern curves corresponding to specific azimuth angles and elevation angles are calculated and drawn in different colors. The antenna gain has been normalized. Furthermore, the variation curves with the azimuth angles and elevation angles are, respectively, shown in Figs. 8 and 9. The change of antenna gain with the angles seems reasonable, and there is a good coincidence degree in the azimuth direction.

Using a large number of measured data from both the WindRAD and the ARC, on the basis of quality inspection and low-quality data elimination, the 3-D antenna pattern is obtained by fitting, which results are given in Fig. 10. Fig. 10(a) and (b) are the 3-D antenna patterns of HH and VV polarizations for C-band, respectively; (c) and (d) are those of HH and VV polarizations for Ku-band, respectively. The shapes of the whole antenna patterns are very similar to the antenna patterns measured in the anechoic chamber before launch. On the one hand, it proves that the calculation method of antenna pattern proposed in this article is correct. On the other hand, it is preliminarily confirmed that the antenna pattern of the WindRAD has not changed obviously due to launch or in-orbit environment.

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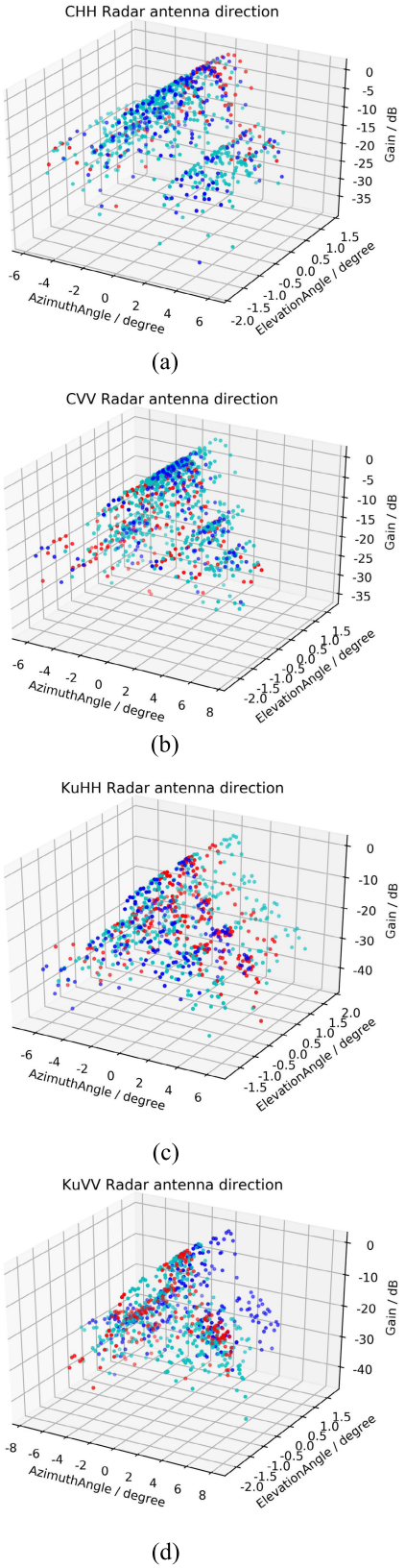


Fig. 7. Antenna pattern calculated by single flyover. (a) C-band, HH polarization. (b) C-band, VV polarization. (c) Ku-band, HH polarization. (d) Ku-band, VV polarization.

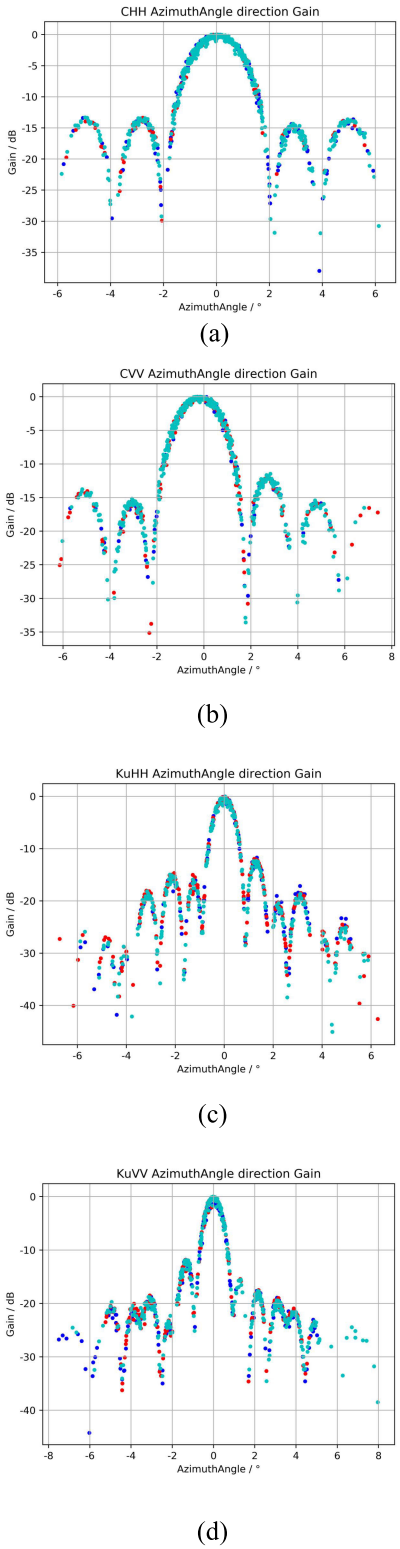
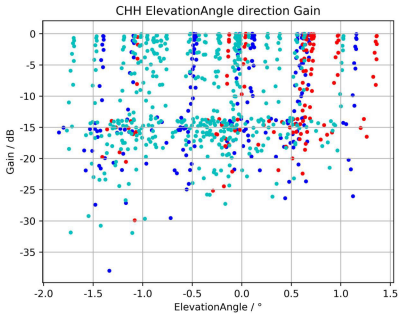


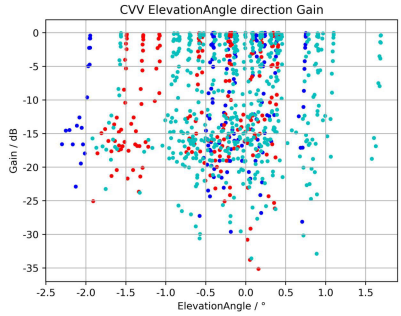
Fig. 8. Variation curve with the azimuth angle. (a) C-band, HH polarization. (b) C-band, VV polarization. (c) Ku-band, HH polarization. (d) Ku-band, VV polarization.

Further comparison is carried out on the difference between the in-orbit antenna pattern and that measured before launch. Fig. 11 shows the differences of each sampling for HH and VV polarizations of C- and Ku-bands. There is a certain degree of jitter in the results of the differences. This might be attributed to several factors. First, the accuracy of pointing calculation could play a role. Inaccuracies in determining the direction and orientation can lead to fluctuations in the results.

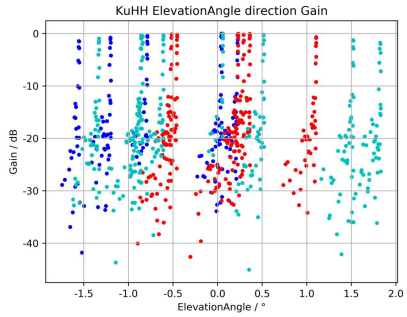
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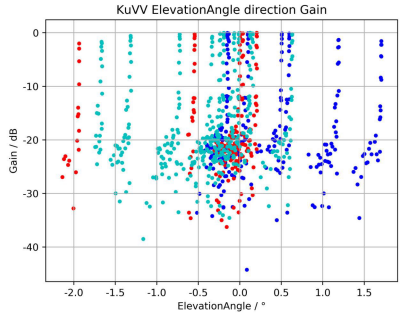
(a)



(b)

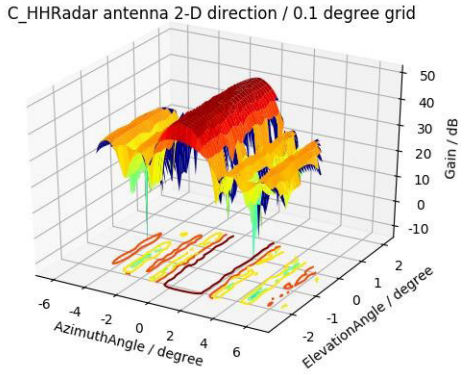


(c)

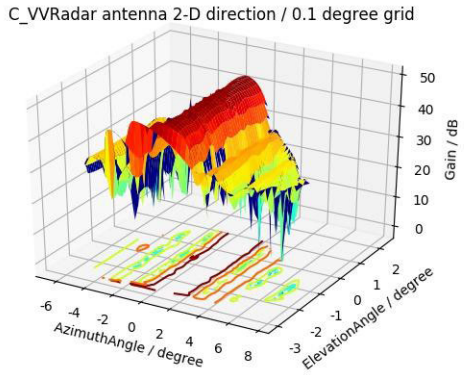


(d)

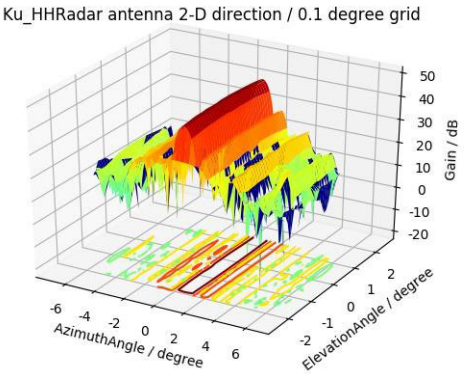
Fig. 9. Variation curve with the elevation angle. (a) C-band, HH polarization. (b) C-band, VV polarization. (c) Ku-band, HH polarization. (d) Ku-band, VV polarization.



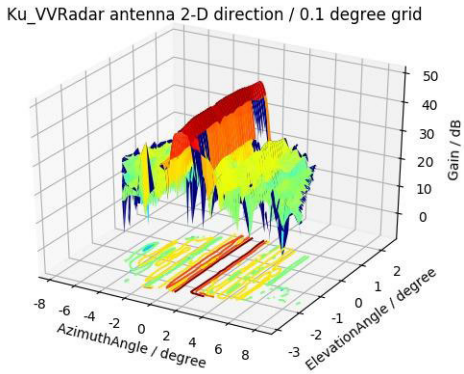
(a)



(b)



(c)



(d)

Fig. 10. Three-dimensional antenna pattern calculated for the WindRAD. (a) C-band, HH polarization. (b) C-band, VV polarization. (c) Ku-band, HH polarization. (d) Ku-band, VV polarization.

Second, weather conditions can also have an impact. Changes in temperature, humidity, and atmospheric pressure can affect the performance of the system and introduce variations in the measured differences. However, generally speaking, these differences are within reasonable ranges. Despite the presence

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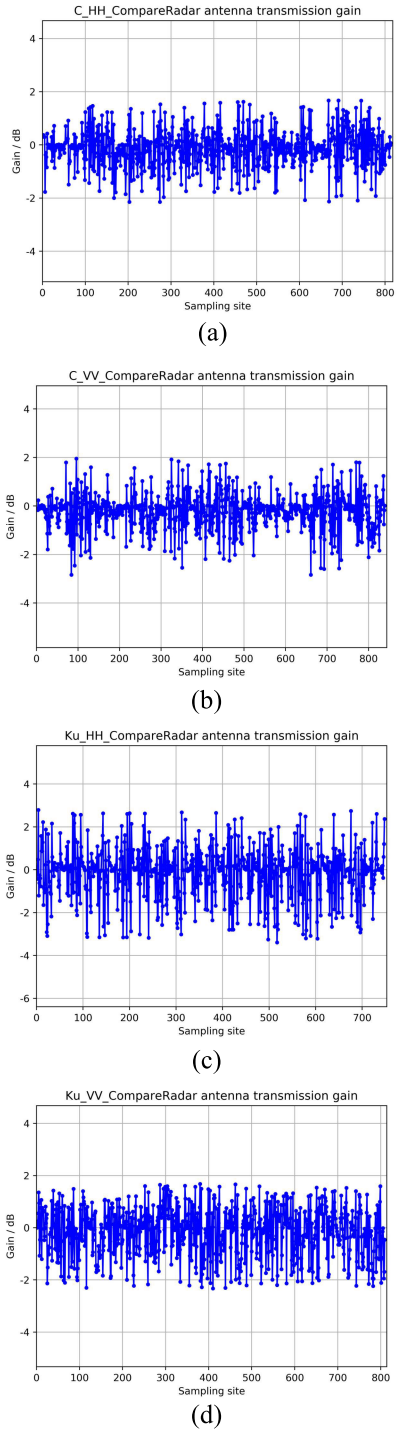


Fig. 11. Difference between the in-orbit antenna pattern and that measured before launch. (a) C-band, HH polarization. (b) C-band, VV polarization. (c) Ku-band, HH polarization. (d) Ku-band, VV polarization.

of jitter, they do not deviate significantly from what would be expected and can be considered acceptable within the context of the overall measurement and analysis.

C. Results of Absolute Calibration Coefficient

Using the method given in Section III, the theoretical NRCS and actual NRCS of the ARC calculated from the observation data of the WindRAD are both obtained, using all the flyover data with adequate signal-to-noise ratio. The received signal of the WindRAD is designed to pass through two channels with different bandwidths simultaneously. Based on the bandwidth ratio and gain ratio of these two channels, the signal-to-noise ratio of the echo can be estimated. During the calculation process, only measurements with the signal-to-noise ratio greater than 26 dB (determined by both the accuracy and the calculation cost) are used to evaluate the radiation measurement accuracy of the WindRAD. The calibration coefficients for the four channels of the WindRAD are listed in Table II. The standard deviations of HH and VV polarizations for C-band are 0.646 and 0.235 dB, respectively. For Ku-band, the standard deviations are 0.760 and 0.595 dB, respectively. The jitter of Ku-band results are bigger than those of C-band. This may be because Ku-band devices are more sensitive and more affected by atmospheric attenuation.

TABLE II
ABSOLUTE CALIBRATION COEFFICIENT RESULTS OF FOUR CHANNELS

Band	Polarization	Standard deviation (dB)
C	HH	0.646
	VV	0.235
Ku	HH	0.760
	VV	0.595

Considering that the stability of observation data is the most critical, because the systematic deviation is easy to be corrected in product retrieval and assimilation applications, we focus on the results of standard deviation. Up to now, the standard deviation is still satisfactory, both for C- and Ku-bands, so no correction has been carried out yet in the calibration process of the WindRAD.

V. CONCLUSION

For the first scatterometer WindRAD onboard FY-3E meteorological satellite, optimal satellite-ground observation mode is designed for the external calibration on account of the functions of the specially developed ARC. The ARC utilizes the frequency shift retransmitting mode to move the calibration signal out of the echo signal frequency range, and also the retransmitting signal power of the ARC can be adjusted to adapt to the observation mode of the WindRAD. Thus, the on-board radar does not need to switch to the external calibration mode, and the routine Earth observation tasks are almost not interrupted. The optimization greatly reduces the complexity of satellite-ground interaction and improves the efficiency of external calibration observation. This allows external calibration to be carried out on a daily basis.

Using the detailed external calibration algorithm proposed in this article, the actual antenna pattern for each channel was calculated and compared with the test result before launch. The in-orbit antenna pattern hadn't changed much, and the key parameters related to the antenna used in WindRAD real-time calibration were strongly confirmed. Also, the absolute calibration coefficient was calculated, although there were many complicated reasons in the satellite-ground link. It can

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examine the overall performance of the instrument, especially considering that internal calibration cannot achieve full link calibration. Based on the calculation results of standard deviation for the biases between the theoretical NRCS and actual NRCS of the ARC, the calibration accuracy is better than 1 dB both for C- and Ku-bands. The results of C-band showed smaller deviation and better stability. WindRAD has the ability to provide high-quality global backscattering measurements, which can be used to retrieve various geophysical parameters and for assimilation applications.

In order to meet the measurement requirements of angular coverage and angular interval of antenna pattern, it takes a long time to complete an external calibration when only a single ground calibration station is used. In addition to the Xilinhot station already built, a second station is being built in Kashgar region, Xinjiang, China. Together with Xilinhot external calibration station, it will serve for high-precision calibration and validation of the WindRAD. The observation data will be further accumulated based on the convenience of the designed observation mode, and the in-orbit stability and accuracy of the WindRAD will be monitored and analyzed continuously.

ACKNOWLEDGMENT

FY-3E/WindRAD data are available from the official data website of the National Satellite Meteorological Center, China Meteorological Administration (NSMC, CMA) (http://data.nsmc.org.cn/PortalSite/Default.aspx?currentculture=en-US).

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